

RT-LAMP Device API Documentation

Overview

This API allows remote control and monitoring of an RT-LAMP thermal cycling device via HTTP requests. The device runs a web server that exposes various endpoints for experiment control, temperature management, fluorescence reading, and configuration.

Connection

Base URL

- **IP Address:** `http://<device-ip>/`
- **mDNS:** `http://<device-mdns>.local/`

The device IP and mDNS name are printed to serial console on successful WiFi connection. By default this is: `http://qbyte.local`. In case mDNS do not work you can use the IP address assigned to the device as a safe fallback option.

If the device do not find the local network it was configured to connect to, it will create a WiFi Access Point named “qByte-Setup”. You can connect to this AP and access the configuration portal at `http://qbyte.local` or `http://192.168.4.1`. In such a portal you can configure a new wifi network for the device to connect to. You can also do this via serial commands, sending ‘wifi “ssid” “password”’ (with “ ” included) via a serial terminal working at 115200 baud rate.

Remember that the USB serial wire connected to a laptop is only required for initial configuration. Once the device is connected to a WiFi network, all interactions should be done wirelessly and the device should be powered by its USB-C 12V charger.

Authentication

Currently, no authentication is required.

API Endpoints

1. Device Status & Information

GET /OnGoing Check if an experiment is currently running.

Response: - 1 - Experiment is running - 0 - No experiment running

Example:

GET `http://device-ip/OnGoing`

GET /free_memory Get available SPIFFS storage space percentage (this is used to store the current protocol data and is erased once a new protocol starts).

Response: - Text: Percentage of free SPIFFS space

Example:

GET `http://device-ip/free_memory`

2. Temperature Control

GET /temp Get current temperature of WELL1 (which is supposed to be a proxy of all wells).

Response: - Text: Temperature value with 4 decimal places (e.g., 25.1234)

Example:

GET http://device-ip/temp

GET /new_temp?degrees=<value> Set new target temperature for the heating wells.

Parameters: - degrees (required): Target temperature in Celsius (0-100°C)

Response: - Success: TARGET TEMPERATURE SET TO <value>°C - Error: [ERROR] TEMPERATURE NOT IN RANGE [0°C-100°C] - Error: [ERROR] TARGET TEMPERATURE REQUIRED

Example:

GET http://device-ip/new_temp?degrees=65

GET /lid_temp Get or set lid temperature.

Parameters: - set (optional): New lid temperature in Celsius

Response: - Without set: Current lid temperature - With set: [OK] LID TEMPERATURE CHANGED TO <value>°C

Examples:

Get current lid temperature

GET http://device-ip/lid_temp

Set lid temperature to 105°C

GET http://device-ip/lid_temp?set=105

GET /lid_diff Get or set lid temperature difference threshold.

Parameters: - set (optional): New lid temperature difference value in Celsius

Response: - Without set: Current lid temperature difference - With set: [OK] LID TEMPERATURE DIFFERENCE CHANGED TO <value>°C

Examples:

Get current difference

GET http://device-ip/lid_diff

Set difference to 5°C

GET http://device-ip/lid_diff?set=5

3. Protocol Execution

GET /Run?degrees=<value> Start a new thermal cycling protocol at specified temperature.

Parameters: - degrees (required): Target temperature in Celsius

Response: - Success: [OK] STARTING NEW PROTOCOL AT <value>°C - Error: [ERROR] TARGET TEMPERATURE REQUIRED

Side Effects: - Resets all variable gains to 0 - Creates/overwrites /last_run.txt file - Sets OnGoing flag to true

Example:

GET http://device-ip/Run?degrees=65

GET /RunMelting Start a melting curve analysis protocol.

Response: - [OK] STARTING NEW MELTING CURVE

Side Effects: - Creates/overwrites /last_run.txt file - Sets OnGoing and OnGoingMelting flags to true - Uses temperature range from melting_range configuration

Example:

GET http://device-ip/RunMelting

GET /Stop Stop the currently running protocol.

Response: - [OK] STOPPING PROTOCOL

Side Effects: - Sets target temperature to 0 - Writes “stop” to /last_run.txt - Clears OnGoing flag

Example:

GET http://device-ip/Stop

GET /manual_cycle Force an immediate thermal cycle and fluorescence reading, bypassing the normal cycle time interval. This is useful when you want to take a reading immediately without waiting for the scheduled cycle time.

Response: - Success: [OK] STARTING MANUAL CYCLE - Error: [ERROR] PROTOCOL NOT RUNNING

Side Effects: - Triggers an immediate cycle regardless of cycle_time setting - Automatically performs fluorescence reading as part of the cycle **Example:**

GET http://device-ip/manual_cycle

4. Fluorescence Reading

GET /ReadFluo Request a fluorescence measurement to be taken. This sets a flag that tells the device to perform a fluorescence reading on the next sensor update cycle. Use this when you want to take a standalone fluorescence measurement without running a full protocol cycle.

Response: - [OK] READING FLUORESCENCE

Side Effects: - Sets `reading_fluorescence` flag to true - Device will perform the reading asynchronously on next sensor update

Note: This is different from `/manual_cycle` which triggers a full thermal cycle. Use `/ReadFluo` for quick fluorescence checks without re running the PID. In reality, both are quite equivalent and just differ slightly in timing, which is useful for debug. Normally use the `manual_cycle` as for protocol runs is safer.

Example:

```
GET http://device-ip/ReadFluo
```

GET /fluo Retrieve the most recent fluorescence measurement values from all 8 channels. This returns cached data from the last reading (triggered by either `/ReadFluo`, `/manual_cycle`, or automatic cycle during a running protocol).

Response: - Text: Comma-separated fluorescence values for all 8 channels

Note: This endpoint only retrieves stored values - it does not trigger a new reading. Call `/ReadFluo` first if you need fresh data.

Example:

```
# Workflow: Request reading, wait briefly, then retrieve values
GET http://device-ip/ReadFluo
# Wait ~1-2 seconds for reading to complete
GET http://device-ip/fluo
# Response example: 1234.5,2345.6,3456.7,4567.8,5678.9,6789.0,7890.1,8901.2
```

5. Calibration

GET /Calibration Start calibration procedure.

Response: - [OK] CALIBRATION STARTED

Side Effects: - Sets `calibrate` flag to true - Calibration may take approximately 1 minute

Example:

```
GET http://device-ip/Calibration
```

GET /readWeights Get calibration weight values.

Response: - Text: Comma-separated list of 8 weight values

Example:

```
GET http://device-ip/readWeights
# Response: 1.234,5.678,9.012,3.456,7.890,1.234,5.678,9.012
```

GET /weights Get calibration weight values (alternative endpoint).

Response: - Text: Comma-separated list of 8 weight values

Example:

```
GET http://device-ip/weights
```

6. Data Retrieval

GET /data_request Get current experiment data.

Response: - Success: Contents of /last_run.txt file - Error: [ERROR] NO PROTOCOL RUNNING (if no protocol is running and last one already retrieved)

Example:

GET http://device-ip/data_request

GET /last_run_request Get data from the last completed run.

Response: - Contents of /last_run.txt file

Example:

GET http://device-ip/last_run_request

7. Configuration Parameters

GET /cycle_time Get or set cycle time in milliseconds.

Parameters: - set (optional): New cycle time in milliseconds

Response: - Without set: Current cycle time value - With set: [OK] CYCLE TIME CHANGED TO <value>ms

Examples:

Get current cycle time

GET http://device-ip/cycle_time

Set cycle time to 30000ms (30 seconds)

GET http://device-ip/cycle_time?set=30000

GET /melting_step Get or set temperature step for melting curve.

Parameters: - set (optional): New melting step in Celsius

Response: - Without set: Current melting step value - With set: [OK] MELTING STEP CHANGED TO <value>°C

Examples:

Get current melting step

GET http://device-ip/melting_step

Set melting step to 0.5°C

GET http://device-ip/melting_step?set=0.5

GET /melting_time Get or set time per step for melting curve.

Parameters: - set (optional): New melting time in milliseconds

Response: - Without set: Current melting time value - With set: [OK] MELTING TIME CHANGED TO <value>ms

Examples:

Get current melting time

GET http://device-ip/melting_time

Set melting time to 5000ms

GET http://device-ip/melting_time?set=5000

GET /melting_range Get or set temperature range for melting curve [start, end].

Parameters: - set (optional): New melting range as “start,end” (e.g., “60,95”)

Response: - Without set: Current range as “start,end” - With set: [OK] MELTING RANGE CHANGED TO <range>°C

Examples:

Get current melting range

GET http://device-ip/melting_range

Response: 60,95

Set melting range from 60°C to 95°C

GET http://device-ip/melting_range?set=60,95

8. Protocol Library Management

GET /protocols Retrieve all saved protocols.

Response: - JSON object containing all saved protocols

Example:

GET http://device-ip/protocols

GET /protocols?save_new=1&name=<protocol_name>&protocol_data=<json_data> Save a new protocol to the library.

Parameters: - save_new: Must be set to 1 - name (required): Protocol name - protocol_data (required): JSON-encoded protocol data

Response: - Success: [OK] Protocol saved successfully - Error: [ERROR] Invalid protocol data format - Error: [ERROR] Protocol name is required - Error: [ERROR] Failed to open protocols file for writing

Example:

GET http://device-ip/protocols?save_new=1&name=MyProtocol&protocol_data={"temp":65,"cycles":40}

GET /protocols?delete=1&name=<protocol_name> Delete a protocol from the library.

Parameters: - delete: Must be set to 1 - name (required): Protocol name to delete

Response: - Success: [OK] Protocol deleted successfully - Error: [ERROR] Protocol not found - Error: [ERROR] Failed to parse existing protocols

Example:

GET http://device-ip/protocols?delete=1&name=MyProtocol

9. Diagnostics

GET /res Get resistance measurement.

Response: - Text: Resistance value

Example:

GET http://device-ip/res

GET /variable_gains Get variable gain values for all 8 channels.

Response: - Text: Comma-separated list of 8 gain values

Example:

GET http://device-ip/variable_gains
Response: 0,0,0,0,0,0,0,0

GET /led_trial Test LED functionality (cycles through all 8 LEDs).

Response: - ok

Side Effects: - Sequentially turns on each LED for 500ms

Example:

GET http://device-ip/led_trial

Common Workflows

Starting an Experiment

1. Check if device is ready

GET http://device-ip/OnGoing

Should return: 0

2. Set desired temperature and start protocol

GET http://device-ip/Run?degrees=65

3. Monitor progress

GET http://device-ip/data_request

Running a Melting Curve

```
# 1. Configure melting parameters (optional)
GET http://device-ip/melting_range?set=60,95
GET http://device-ip/melting_step?set=0.5
GET http://device-ip/melting_time?set=5000
```

```
# 2. Start melting curve
GET http://device-ip/RunMelting
```

```
# 3. Monitor progress
GET http://device-ip/data_request
```

Stopping an Experiment

```
GET http://device-ip/Stop
```

Reading Temperature and Fluorescence

```
# Get current temperature
GET http://device-ip/temp
```

```
# Trigger fluorescence reading
GET http://device-ip/ReadFluo
```

```
# Wait a moment, then retrieve values
GET http://device-ip/fluo
```

Notes

1. **All endpoints use HTTP GET requests** - No POST methods are currently used in the implementation
 2. **Configuration changes are persisted** - Settings like cycle_time, lid_temp, melting parameters are saved to device storage
 3. **Protocol data is stored in SPIFFS** - The /last_run.txt file contains experiment data
 4. **No concurrent experiments** - Only one protocol can run at a time
 5. **Response format** - Most responses are plain text, protocol library uses JSON format but sent in plain text
 6. **Temperature units** - All temperatures are in Celsius
 7. **Time units** - Cycle and melting times are in milliseconds
-

Error Handling

The API returns HTTP 200 status for most requests, with error messages in the response body starting with [ERROR]. Success messages start with [OK].

Common error patterns: - Missing required parameters: [ERROR] TARGET TEMPERATURE REQUIRED - Invalid ranges: [ERROR] TEMPERATURE NOT IN RANGE [0°C-100°C] - Invalid state: [ERROR] NO PROTOCOL RUNNING - File system errors: [ERROR] Failed to open protocols file for writing
